

ESP32 Engine Controller - Quick Guide

(BETA)

Hardware Connections

Component	ESP32 Pin	Notes
ESC Signal	GPIO 18	Yellow wire from ESC
L298N EN	GPIO 23	Speed control (PWM)
L298N IN1	GPIO 22	Direction pin 1
L298N IN2	GPIO 21	Direction pin 2
Spark Relay	GPIO 26	Controls relay to spark plug

L298N Motor Driver Wiring:

- **12V Power** → L298N 12V input
- **GND** → Common ground (ESP32 + L298N + 12V supply)
- **Motor** → L298N OUT1 & OUT2
- **EN, IN1, IN2** → ESP32 as shown above

Blynk App Setup

Virtual Pin	Widget	Purpose
V1	Slider (0-100)	Manual ESC control
V2	Slider (0-100)	Manual fuel pump
V3	Button	Manual spark on/off
V4	LED	Manual mode indicator
V5	LED	Auto mode indicator
V10	Slider (0-100)	Auto mode ESC target
V11	Slider (200-60000)	Auto spark period (milliseconds)
V12	Slider (0-100)	Auto mode pump target
V13	Button	Switch Manual ↔ Auto

How It Works

Manual Mode (Default)

You're the boss. Control everything yourself:

- Slide V1 to control motor speed (0-100%)
- Slide V2 to control fuel pump (0-100%)
- Press V3 to turn spark plug on/off
- For testing on manual operation

Auto Mode (Press V13 to activate)

The system runs a pre-programmed startup sequence automatically:

1. **Gentle Startup:** Ramps ESC through 10% → 20% → 30% → 50%
2. **Noise Check:** Holds at 50% for 10 seconds (listen for issues)
3. **Full Power:** Increases to your target ESC% (default 80%)
4. **Stabilize:** Waits 10 seconds at target speed
5. **Ignition:** Starts spark plug cycling (default every 2 seconds)
6. **Fuel Injection:** Starts fuel pump at target% (default 50%)
7. **Running:** System keeps running until you switch back to manual

Before starting Auto Mode, set your targets:

- V10: How fast you want the motor (default 80%)
 - V11: How often spark fires (2000ms = spark toggles every 1 second)
 - V12: How much fuel to inject (default 50%)
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Quick Start

1. **Upload code** with your WiFi credentials and Blynk token
 2. **Wire everything** according to the table above
 3. **Power up** - System starts in Manual Mode
 4. **Test manually** - Try sliders V1, V2, and button V3
 5. **Set auto parameters** - Adjust V10, V11, V12 if needed
 6. **Go auto** - Press V13 to start automatic sequence
 7. **Emergency stop** - Press V13 again to return to Manual (everything stops)
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Safety Notes

- **Always start in Manual Mode** to test each component
 - **Emergency stop:** Switch back to Manual Mode - everything shuts off immediately
 - **Auto Mode only starts** when you're already in Manual Mode (safety feature)
 - System resets to safe state (everything OFF) when switching modes
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Troubleshooting

Problem	Solution
Motor not responding	Check ESC calibration (runs at startup)
Pump not working	Check L298N wiring and 12V power
Can't switch to Auto	Must be in Manual Mode first
Spark not firing	Check relay wiring to GPIO 26
Blynk disconnected	Check WiFi credentials in code
